ANSI/IES LM-80-21

Illuminating
ENGINEERING SOCIETY

APPROVED METHOD: MEASURING MAINTENANCE OF LIGHT OUTPUT CHARACTERISTICS OF SOLID-STATE LIGHT SOURCES

AN AMERICAN NATIONAL STANDARD

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has been approved by IES.
Suggestions for revisions
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1.0 Introduction and Scope

1.1 Introduction

The output in optical radiation from solid-state light sources such as LEDs and laser diodes eventually decreases over time. This characteristic of declining output without sudden and permanent failure creates a risk that products incorporating a solid-state light source, that is operating near end of life may be performing outside the products' specifications, or outside required codes, standard practices, or regulations. These solid-state light sources may also undergo gradual shifts in the emitted spectrum over time that may result in unacceptable appearance, color rendering, efficacy, or efficiency.

Product characteristics obtained using this procedure are measured under controlled conditions that may allow direct comparison of results obtained at different laboratories. The resulting data may be used for direct evaluation and comparison of solid-state light source components, and they may be utilized in models that project long-term changes in performance characteristics.

1.2 Scope

This Approved Method describes the procedures by which solid-state light sources, such as LED packages, arrays and modules; or laser diode packages, arrays and modules may be tested for flux maintenance over time, including luminous flux, radiant flux, and photon flux maintenance. This document also provides methods for measurement of spectrum-dependent characteristic maintenance, including changes in chromaticity coordinates, peak wavelength, dominant wavelength, centroid wavelength, and spectral power distribution over time, when carried out under controlled environmental and operational conditions. For the purposes of this document, *solid-state light sources* include ultraviolet, visible, and infrared sources emitting optical radiation in the range of 200 nm to 2,000 nm.

This Approved Method does not cover lamps, light engines, or luminaires and does not provide guidance regarding predictive estimations or extrapolation for the maintenance characteristics beyond the duration of the actual measurements.

2.0 Normative References

2.1 ANSI/IES LS-1-20

Illuminating Engineering Society. Recommended Practice: Nomenclature and Definitions for Illuminating Engineering. New York: IES; 2020. Online: www.ies.org/standards/definitions/ (Accessed 2020 Dec 7).

2.2 ANSI/IES TM-27-20

Illuminating Engineering Society. Technical Memorandum: IES Standard Format for the Electronic Transfer of Spectral Data. New York: IES; 2020.

2.3 ASTM Standard E230/E23M-112

ASTM International. ASTM E230/E230M-17, Standard Specification and Temperature-Electromotive Force (EMF) Tables for Standardized Thermocouples. West Conshohocken, PA: ASTM International; 2017.

3.0 Definitions

Definitions found in this section are for the purpose of this document. Definitions not found in this section may be found in *ANSI/IES LS-1-20, Recommended Practice: Nomenclature and Definitions for Illuminating Engineering* (see **Section 2.1).**

3.1 air temperature (T_A)

The temperature of the air surrounding the DUT (device under test) during the maintenance test.

3.2 centroid wavelength (λ_c)

The wavelength calculated as the weighted center of a spectral distribution according to:

$$\lambda_c = \frac{\int_{\lambda_1}^{\lambda_2} \lambda \bullet \Phi_\lambda(\lambda)\, d\lambda}{\int_{\lambda_1}^{\lambda_2} \Phi_\lambda(\lambda)\, d\lambda},$$

where:
λ is the wavelength
$\Phi_\lambda(\lambda)$ is the spectral distribution
λ_1 and λ_2 are the limits for the wavelength range of $\Phi_\lambda(\lambda)$.

Units: nanometers (nm).

Note: ANSI/LS-1-20 (refer to **Section 2.1**) defines *centroid wavelength* for monochromatic light. For the purposes of this document, the term is not so restricted.

3.3 device under test (DUT)

A solid-state light source component, such as an LED package, LED array and module, laser diode package, or laser diode array and module, that is undergoing the maintenance test.

3.4 dominant wavelength (of a color stimulus) (λ_d)

The wavelength of the monochromatic stimulus that, when additively mixed in suitable proportions with the specified achromatic stimulus, matches the color stimulus considered in the CIE 1931 (*x, y*) chromaticity diagram.

Unit: nanometer (nm)

Note 1: In the case of a purple stimulus, the dominant wavelength is replaced by the complementary wavelength.

Note 2: In this document, Illuminant E (equi-energy spectrum) is the achromatic stimulus.

3.5 drive level

The nominal external voltage or current applied to a DUT during the maintenance test or during an optical or electrical measurement. Drive level is specified in amperes for DC constant-current drive, volts for DC constant-voltage drive, or RMS volts for AC regulated voltage.

3.6 luminous flux maintenance

The remaining luminous flux output as a function of elapsed operating time.

Note: Luminous flux output is typically measured at selected operating time points. The remaining luminous flux output is typically expressed as a percentage of the initial luminous flux output. In other documents, *lumen maintenance* is often used. Luminous flux maintenance is the converse of *luminous flux depreciation* (or *lumen depreciation*).

3.7 maintenance test

The continuous operation of the DUT when it is energized under specific steady-state electrical and environmental conditions.

3.8 measurement interval

The elapsed time between two sequential optical and electrical measurements.

Units: hours.

3.9 optical radiation

Optical radiation is defined in ANSI/IES LS-1-20 (refer to **Section 2.1**) as "electromagnetic radiation having wavelengths between approximately 100 nm and 1 mm."

For the purposes of this document, the term *optical radiation* includes wavelengths within the range of 200 nm to 2,000 nm (2 µm).

3.10 peak wavelength (λ_p)

The wavelength at the maximum amplitude of the spectral distribution.

Note: If there are two or more peaks in the spectral distribution, a local peak wavelength for a portion of the spectral distribution may be reported, in which case the portion should be defined by wavelength range or other term (e.g., pump emission).

Units: nanometers (nm).

3.11 photon flux (Φ_p)

The photon flux emitted by a DUT in the wavelength interval from λ_1 to λ_2 is:

$$\Phi_p = \int_{\lambda_1}^{\lambda_2} \frac{\lambda}{Nhc} \Phi_\lambda(\lambda)\, d\lambda ,$$

where:
N is Avogadro's number
h is Planck's constant
c is the speed of light
$\Phi_\lambda(\lambda)$ is the spectral radiant flux, measured in W/nm
λ_1 and λ_2 are the limits for the entire wavelength range of $\Phi_\lambda(\lambda)$, or a specified wavelength range

The units for photon flux, Φ_p, are moles per second (mol/s), usually expressed as micromoles per second (µmol/s).

3.12 photon flux maintenance

The remaining photon flux output) as a function of elapsed operating time.

Note: Photon flux output is typically measured at selected operating time points. The remaining photon flux output is typically expressed as a percentage of the initial photon flux output.

3.13 radiant flux (Φ_e)

Radiant flux is generally defined in ANSI/IES LS-1-20 (see **Section 2.1**). For the purposes of this document, *radiant flux* is calculated as the integral of spectral radiant flux $\Phi_\lambda(\lambda)$ over a wavelength range:

$$\Phi_\mathrm{e} = \int_{\lambda_1}^{\lambda_2} \Phi_\lambda(\lambda)\, \mathrm{d}\lambda ,$$

where λ_1 and λ_2 are the limits for the entire wavelength range of $\Phi_\lambda(\lambda)$, or for a specified wavelength range.

Units: watts.

3.14 radiant flux maintenance

The remaining radiant flux output as a function of elapsed operating time.

Note: Radiant flux output is typically measured at selected operating time points. The remaining radiant flux output is typically expressed as a percentage of the initial radiant flux output.

3.15 temperature measurement point (TMP)

A light-source-manufacturer-designated location on the solid-state light source package, or a nearby point external to the package, at which the temperature associated with a DUT's performance is measured.

3.16 TMP temperature

The measured temperature of the temperature measurement point (TMP) for the device under test (DUT). In previous versions of ANSI/IES LM-80, TMP temperature was called case temperature.

4.0 Physical and Environmental Conditions

4.1 General

It is recommended laboratory practice that the storage and testing of DUTs should be undertaken in a relatively clean environment. Prior to operation, DUTs should be cleaned to eliminate handling marks, and the manufacturer's handling instructions should be observed; e.g., electro-static discharge (ESD) instructions. DUT testing chambers or test environments should not release volatile organic compounds, halogen or sulfur compounds, or other contaminants that can interact with silicone, lead frames, or other components in a DUT. All portions of the maintenance test shall be conducted in an air environment.

4.2 Humidity

During the maintenance test, the relative humidity (RH) shall be maintained either at less than 65% or at another pre-defined relative humidity (RH) level (for specific applications), within a tolerance of ±5%, throughout the maintenance test. The chosen humidity test condition shall be reported.

4.3 Air Temperature and Air Movement

During the maintenance test, the impact of airflow (e.g., active or passive convection) on the temperature of the TMP should be minimized. Air temperature, T_A, shall be maintained at a temperature that is greater than the nominal test TMP temperature minus 5 °C (e.g., if nominal TMP temperature = 55 °C, then T_A shall be 50 °C or greater). T_A should be monitored within the test chamber or environment during the maintenance test. T_A should be measured at a location chosen to best represent the surrounding air without introducing inaccuracies (e.g., due to light absorption).

Note: Maintenance tests conducted in a chamber may utilize passive or active convection methods to maintain the air temperature.

4.4 Temperature Measurement Equipment or System

The temperature measurement equipment or system shall have an expanded uncertainty ($k = 2$) of less than 2.5 °C. If thermocouples are used, the wire shall comply with ASTM E230 Table 1 "Special Limits" (see **Section 2.3**). Temperature sensor elements shall be shielded from direct DUT optical radiation.

5.0 Electrical Conditions

5.1 DUT Drivers

DUTs shall be operated using external drivers. Electric power provided shall be one of the following:

- DC constant current
- Pulse-width modulated (PWM) current
- DC constant voltage
- AC regulated voltage

The input power type should be chosen to match the DUT's primary mode of operation specified by the manufacturer (e.g., constant current, constant voltage, or AC voltage). Drive power shall adhere to the conditions described in **Sections 5.1.1** through **5.1.4**, according to the type of drive employed.

5.1.1 DC Constant Current Drive. For this condition, DUTs shall be driven individually with dedicated drivers, or in series circuits with a constant current. At optical and electrical measurement intervals, DUT forward voltage shall be measured and reported.

The forward current shall be regulated to within ±3% of the nominal value during the maintenance test. Forward current regulation during optical and electrical measurements shall be within ±0.5% of the nominal value. Peak-to-peak current ripple shall not exceed 3% of the nominal value at any time.

5.1.2 Pulse Width Modulated (PWM) Current Drive. For this condition, DUTs shall be driven individually with dedicated drivers, or in series circuits with a PWM current. At optical and electrical measurement intervals, DUT forward voltage shall be measured and reported.

The forward current shall be regulated to within ±3% of the nominal value during the maintenance test. Forward current regulation during optical and electrical measurements shall be within ±1% of the nominal value. PWM current transitions (rising and falling current edges) shall occur within 1% of nominal on/off transition times during optical and electrical measurement, and within 3% of nominal on/off transition times for the maintenance test. The cabling between the drivers and DUTs should be designed to minimize current errors due to cabling losses. For example, an inappropriately designed cable could result in current bypassing the DUTs by flowing through parallel cabling capacitance.

5.1.3 DC Constant Voltage Drive. For this condition, DUTs shall be driven individually with dedicated drivers, or in parallel circuits. External current regulating components such as resistors may be used per the manufacturer's recommendation. At optical and electrical measurement intervals, DUT forward current shall be measured and reported.

Forward voltage shall be regulated to within ±1% of the nominal value during the maintenance test, and within ±0.5% of the nominal value during optical and electrical measurements.

5.1.4 AC Regulated Voltage Drive. For this condition, DUTs shall be driven individually with dedicated drivers, or in parallel circuits. The AC drive shall operate at the same frequency throughout the maintenance test. External current regulating components such as resistors may be used per the manufacturer's recommendation. At optical and electrical measurement intervals, DUT root-mean-square (RMS) current shall be measured and reported.

The AC RMS voltage shall be regulated to within ±1.5% of the nominal value during the maintenance test, and within ±0.5% of the nominal value during optical and electrical measurements. Voltage total harmonic distortion of the AC voltage source shall not exceed ±3%, and the AC frequency shall be maintained within ±0.5% during maintenance testing.

5.2 Drive Level Value

The drive level value used for the maintenance test (DC current, DC voltage, or AC RMS voltage and frequency) shall represent the manufacturer's expectation for users' applications, and it should be within the DUT's recommended operating range. Ideally, the value should match that used for the manufacturer's datasheet flux rating. The drive level value and any other drive parameters (e.g., PWM parameters, AC frequency) shall be reported.

6.0 Optical and Electrical Measurement Procedures

Prior to commencing the maintenance test, and at each measurement interval, optical and electrical measurements for the DUTs shall be performed per the following procedures.

6.1 DUT Optical and Electrical Measurement System and Measurement Method

The measurement system may include an integrating sphere, hemisphere, or goniometer. The system's optical radiation measuring device, including photometer, radiometer, or spectrometer, should be chosen to maximize measurement accuracy and repeatability for a particular parameter of interest. For example, when measuring UV DUT with narrow wavelength bandwidth, high-quality photodiode setups are the preferred type of detectors.

For integrating sphere system, the measurement of flux and spectrum-dependent characteristics shall be made using a calibrated measurement system that samples at least 98% of the total DUT optical output. This requirement ensures that the optical output at high angles is captured, which is important for monitoring changes in spectrum-dependent characteristics. For goniometer systems, the angular scanning resolution shall be fine enough to accurately characterize the DUT.

Optical and electrical measurements should be performed using a measurement method that optimizes the repeatability of the measurements. ANSI/IES LM-85-20 can be referenced for developing an appropriate method.[1]

6.2 Equipment Calibration and Measurement Corrections

The measurement system and equipment calibration shall be in accordance with the manufacturer's specifications. To minimize errors due to differences in calibration references, the same optical calibration reference should be used throughout the maintenance test. Operating time on this calibration reference should be minimized to improve long-term measurement system reproducibility.

To reduce errors due to changes in the measurement system responsivity (e.g., sphere drift), stable solid-state light sources ("monitor DUTs"), not part of the maintenance test, may also be measured at 0 hours and the same interval as the DUTs undergoing the maintenance test. The monitor DUT measurements can then be used to correct for changes in the measurement system's responsivity. Any corrections made using monitor DUTs shall be documented in the test report.

For sphere-based measurements, a self-absorption correction shall be made at each measurement interval for each DUT.

6.3 Flux Measurements

The maintenance test includes initial and subsequent measurements of luminous flux, radiant flux, or photon flux. The initial (at 0 hours) measurement value of radiant flux, photon flux, or luminous flux for each DUT shall be reported in SI units. Subsequent flux measurements shall be normalized to a value of 1 (100%) at 0 hours. For such normalized measurements, many systematic error components will cancel out, assuming the same measurement setup is used with no changes during maintenance testing of a DUT.

6.4 Measurement of Spectrum-Dependent Characteristics

The maintenance test also includes initial and subsequent measurement of spectrum-dependent characteristics such as chromaticity, peak wavelength, dominant wavelength, centroid wavelength, and spectral power distribution (SPD).

6.4.1 Chromaticity. If applicable, the initial (at 0 hours) and subsequent measurement data of chromaticity for each DUT shall be reported in CIE 1976 (u', v') coordinates, CIE 1931 (x, y) coordinates, or both.

In addition, the change in chromaticity at a measurement interval at t hours may be reported as the geometric distance between the chromaticity at 0 hours (u'_0, v'_0) and the chromaticity at t hours (u'_t, v'_t):

$$\Delta u' = u'_t - u'_0$$

$$\Delta v' = v'_t - v'_0$$

$$\Delta u'v' = \sqrt{(\Delta u')^2 + (\Delta v')^2}$$

6.4.2 Peak Wavelength. If applicable, the initial (at 0 hours) and subsequent measurement data of the peak wavelength for each DUT shall be reported. If the peak wavelength is not calculated as the wavelength of the data point of maximum intensity, the method of calculation for peak wavelength shall be specified.

In addition, the change in peak wavelength at a measurement interval at t hours may be reported.

6.4.3 Dominant Wavelength. If applicable, the initial (at 0 hours) and subsequent measurement data of the dominant wavelength for each DUT shall be reported.

In addition, the change in dominant wavelength at a measurement interval at t hours may be reported.

6.4.4 Centroid Wavelength. If applicable, the initial (at 0 hours) and subsequent measurement data of the centroid wavelength for each DUT shall be reported.

In addition, the change in centroid wavelength at a measurement interval at t hours may be reported.

6.4.5 Spectral Power Distribution. If applicable, the initial (at 0 hours) and subsequent measurement data for each DUT SPD shall be reported per the electronic data format (SPDX) specified in ANSI/IES TM-27 (see **Section 2.2).**

6.5 DUT Measurement Temperature Condition

The same DUT temperature condition shall be used for all DUT measurements. A condition of 25 °C is commonly used in industry. To facilitate comparison of measurement data, the temperature condition and the ambient temperature measurement location shall be reported. The measurement location may be a laboratory central monitoring point, such as a room ambient air temperature monitor, a sphere ambient air temperature probe, the temperature of an actively controlled heat sink, or any other point that can be correlated to the DUT selected temperature condition. The temperature at the ambient measurement location shall be within ±2 °C of the measurement temperature condition prior to starting any measurement. Improved long-term measurement consistency can be achieved by maintaining a temperature stability within ±0.2 °C.

6.6 DUT Measurement Drive Level

Optical and electrical measurements shall be made using the maintenance drive level.

7.0 Maintenance Test Preparation

7.1 DUT Identification and Tracking

Each DUT sample that undergoes a maintenance test shall be identified and tracked throughout the test. Extra DUT samples may be useful in the event some samples are determined to be failures under the guidelines listed in **Section 8.3**. All samples tested shall be included in the report.

7.2 Seasoning or Aging

No seasoning or aging shall be performed beyond production seasoning done by the manufacturer for the product prior to its release to end-users. If production seasoning is performed on the DUT sample set, the duration and conditions shall be reported.

7.3 Timekeeping

Accurate recording of elapsed operating time is critical. Measurement intervals shall be timed with computer-based timers, an elapsed time meter, video monitoring, current monitoring, or another monitoring device. A timer shall be associated with a particular test position and shall accumulate time only when the installed DUTs are energized. In the event of a power failure, monitoring devices shall not accumulate time. Timing uncertainty, including time in which the DUTs are energized but not yet within the DUT TMP temperature limits (see **Section 8.1**), shall not exceed 0.5% of the measurement interval.

8.0 Maintenance Test Procedures

A maintenance test shall consist of the initial measurement and at least one additional measurement. Maintenance test data are often used for extrapolating lifetime claims for solid-state light sources. Some methods of extrapolation may allow interpolation between test temperatures and drive levels. These extrapolation methods may also have interval spacing, sample selection, seasoning, and other constraints. If the maintenance test data are intended to be used for extrapolation, the interpolation and extrapolation requirements in other documents (e.g., ANSI/IES TM-21,[2] ANSI/IES TM-28,[3] ANSI/IES TM-35[4]) should be reviewed to ensure utility of the maintenance test data.

8.1 DUT TMP Temperatures

DUTs shall be driven during the maintenance test at a minimum of two TMP temperatures. For solid-state light sources intended for general lighting, at least one of the TMP temperatures should be 55 °C or 85 °C. These TMP temperatures are commonly used for industry testing to support direct product comparisons of testing results.

During the maintenance test, the DUTs' TMP temperature, shall be maintained at a temperature no lower than the nominal test TMP temperature minus 2 °C (e.g., if nominal TMP temperature = 55 °C, then the measured TMP temperature shall be 53 °C or higher). TMP temperatures shall be monitored during the maintenance test using one of the monitoring methods listed in **Annex A.**

8.2 Maintenance Test Duration and Measurement Interval

The maintenance test duration and measurement intervals shall be based on, and consistent with, the design operating life; the intent of the test, including evidence of compliance with the regulations requested; and planned analysis of the data (e.g., data for predictive luminous flux maintenance modeling, or data for actual luminous flux maintenance validation).

8.3 Recording DUT Failures

Checking for DUT failures either by visual observation or by automatic monitoring, shall be done at every optical and electrical measurement interval. A DUT shall be declared a failure if it sustains mechanical damage affecting performance due to mishandling or if the flux parameter of primary interest (e.g., luminous flux, photon flux, or radiant flux) decreases by 90%, to 10% of the initial flux value or lower). A DUT shall also be declared a failure if its operating voltage (for current-driven DUTs) or current (for voltage-driven DUTs) changes by more than 50% during the maintenance test. Failed DUTs shall be reported. Failed DUT optical and electrical measurement data shall be excluded from the report (see **Section 9.0**). (*Note:* It is recognized that failures may be defined differently in safety documents or in other standards or regulatory documents.)

9.0 Test Report

The report shall list all pertinent data concerning conditions of testing, type of equipment, and types and quantities of DUTs. The items listed in **Sections 9.1** through **9.8** shall be reported.

9.1 Administrative Information

1. Testing agency identification
2. Report issue date
3. Testing start date
4. Testing completion date

9.2 DUT Identification

1. DUT manufacturer's name
2. DUT identification; e.g., model number
3. Description of DUT, including whether the DUT is an LED package, array, or module

4. Number of DUTs (sample size) tested
5. If applicable, production seasoning duration, per **Section 7.2**

9.3 Maintenance Test Conditions

1. TMP location, including a DUT image with scale reference showing temperature sensor attachment in sufficient detail that it can be replicated
2. Nominal TMP temperature(s), with expected range of variation
3. Air temperature(s), with expected range of variation
4. Description of air movement
5. Chosen relative humidity level, per **Section 4.2**
6. Drive type, drive level, and any other drive parameters (e.g., PWM parameters, AC frequency) for DUTs

9.4 Test Equipment

1. Description of testing equipment
2. External drive regulating components, if used

9.5 Maintenance Test Duration

Recorded in units of hours; may be rounded to the nearest hour.

9.6 Measurement Intervals

Recorded in units of hours; may be rounded to the nearest hour.

9.7 Failed DUTs

1. Number of DUT failures, per **Section 8.3**
2. Approximate time of each failure observed
3. Type of failure(s) (i.e., mechanical, electrical, or optical)

9.8 Working DUTs: Measured Optical and Electrical Results

1. Initial and subsequent luminous flux, or radiant flux, or photon flux; subsequent measurements shall be reported normalized to the initial value format and reported to the nearest hundredth of a percent
2. Initial and subsequent chromaticity coordinates, or dominant wavelength, or peak wavelength, or centroid wavelength
   a. The peak wavelength may be the peak over the entire distribution or over a localized wavelength range. If a localized wavelength range is evaluated, the limits of the range shall be reported.
   b. For centroid wavelength, λ_1 and λ_2 should be chosen to minimize noise contribution without omitting signal from the DUT. Wavelengths λ_1 and λ_2 shall be reported.
3. Initial and subsequent DUT forward voltage, DUT forward current, or RMS current, per **Section 5**
4. For each parameter in items 1 and 2 that is reported, the average value, median value, standard deviation, and minimum and maximum value for all the DUTs in a given test condition at each measurement interval
5. DUT measurement temperature condition, per **Section 6.5**
6. A description of the optical measurement method and the specific method variant used; any corrections made, such as sphere drift corrections or temperature corrections, shall be included

9.9 Optional Items

1. Statement of uncertainties (if required)
2. Change in chromaticity, as described in **Section 6.4.1**
3. DUT sample selection method, (e.g., random sampling, first production reel)
4. Spectral power distribution (SPD) formatted as SPDX data files, as described in **Section 6.4.5**

Annex A – Methods for Monitoring of TMP Temperatures

TMP temperature monitoring verifies that all DUTs are being operated within the temperature requirements of **Section 8.1**. Ideally, each individual DUT should be monitored using a separate temperature sensor. For maintenance testing on a small scale, direct measurement of DUT TMP temperature is practical. If directly measured, the temperature probe shall be attached at the manufacturer-designated TMP temperature measurement point. For larger-scale testing, usually beyond 10 DUTs, it becomes very challenging to properly attach, shield, and monitor the thermocouples or other sensors needed for direct measurements. If direct measurement of each DUT in a sample set is not practical, one of the methods below may be used. Regardless of the method used, the minimum DUT TMP temperature in a sample set shall be used for the test report.

A.1 Thermal Imaging Camera and Sensor

In this method, DUT temperatures shall be measured using a thermal imaging camera to identify the coldest DUT. This measurement may be made with some adjustments to the testing environment or chamber—such as opening a door—provided that the adjustments do not significantly alter the relative temperatures of the DUTs. As it is difficult to perfectly specify the emissivity, the temperatures obtained shall be treated as relative, rather than absolute. The DUT with the lowest temperature shall then be identified and instrumented with a thermocouple or other temperature sensor. The TMP temperature of this DUT shall then be monitored during the operating interval. This temperature shall be used to represent the entire sample.

A.2 Thermocouples or Other Sensors

In this method, a subset of the DUTs shall be chosen and instrumented with thermocouples or other sensors to determine the relative TMP temperatures of the DUTs. As in the thermal imaging method (see **Section A.1**), the DUT with the lowest TMP temperature shall be identified, and then the remaining thermocouples or other sensors may be removed. The measurements from the lowest TMP temperature DUT shall then be used to represent the entire sample.

A.3 Reference Thermal Sensor

In this method, a reference thermal sensor shall be mounted in close proximity to the DUTs, in a location of a fixed thermal resistance from the DUTs. The reference thermal sensor may be, for example, a resistive temperature detector (RTD) soldered to the DUT circuit card. It shall be attached permanently and reliably to give repeatable, low-uncertainty measurements throughout the maintenance test. To determine the thermal resistance from the sensor to the coldest DUT, the coldest DUT shall first be identified using one of the methods specified in **Section A.1** or **Section A.2**. Once the coldest DUT has been identified, the thermal resistance from this DUT to the reference sensor shall then be measured using a thermocouple or other sensor directly attached to the DUT TMP. After the thermal resistance has been established, the reference thermal sensor shall be monitored during the maintenance test, and the sum of the reference thermal sensor temperature and the DUT TMP temperature rise due to the thermal resistance shall be reported as the minimum DUT TMP temperature for the entire sample.

Annex B – Chip-on-Board (COB) Devices

Chip-on-board LEDs (COBs) offer unique challenges when performing maintenance testing. During LM-80 testing, electrical connections to a COB are made on the top of the package, as shown in **Figure B-1**. The temperature of the COB is typically measured at a TMP location, which is also located on the top of the package. For white COBs, the array of LED dice—covered with a silicone and phosphor mixture—are within the light-emitting surface (LES) area of the package. It is not unusual for dozens of small LED dice to be within the LES. As with other white LEDs, white light is produced by wavelength down-conversion of blue photons emitted by the LED dice that strike phosphor particles.

Unlike the LED package types that are soldered onto PC boards, COB packages do not utilize surface-mount technology (SMT) for attachment to an underlying heat

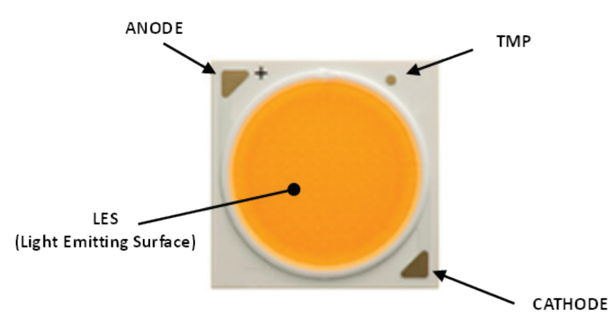

Figure B-1. Typical COB package.

sink. Rather, COB packages are typically mechanically attached to a board and/or a heat sink with screws and a holder that fits over the COB, as shown in **Figure B-2.**

Figure B-2. Holder and COB Package.

When operating a COB in a chamber during maintenance testing, the TMP temperature is maintained at the same temperature as the surrounding air. For example, during testing at 85 °C, the air temperature and TMP temperature are both kept at a nominal 85 °C. As with all LEDs, during operation of the product, the temperature of the LED dice will be higher than that of the surrounding environment. This heat is then transferred from the LED dice to the underlying board or heat sink, where it is dissipated. For SMT packages, this heat transfer is accomplished via the solder joint between the LED package and the PCB. For COBs, no solder is used to attach the COB to the board. Thus, the use of a thermal interface material (e.g., thermal grease, thermal adhesives, or thermally conductive silicone pads) is critical to effectively transfer the heat away from the COB package. Similarly, if a COB holder is being used, it is important that it be properly attached to achieve good surface contact between the backside of the package and the thermal interface material. (Refer to the holder manufacturer's technical specifications when selecting and using such products.)

Improper use of a holder or inadequate application of a thermal interface material can result in unexpectedly high LES temperatures within the package during operation. This can cause poor luminous flux maintenance, damage to the LES silicone, and complete failure of the COB.

For those designing COBs into a lighting product, it is critical that similar care be exercised in the construction of the lighting product. If proper thermal management is not employed in the construction of the product, the product's performance over time may not match performance expectations based on the maintenance test data. In these types of situations, it is possible for the TMP of the COB to be at a satisfactory level while the LES temperature is unacceptably high, resulting in poor lighting product performance. Guidance from the COB manufacturer is important in order to prevent such issues.

Annex C – Additional Laboratory Best Practices

Solid-state device maintenance testing encompasses many disciplines, including heat transfer, electronic engineering, and optical measurement science. The body of this standard provides requirements and suggested practices in these areas. Successful LM-80 laboratories adhere to these requirements as well as to the suggestions. In addition, successful labs have found other practices to be useful:

- **Ensure that the atmosphere within the thermal chambers is free from chemicals that could react with DUTs.** For example, many LED packages employ a silver mirror to reflect light. If sulfur compounds are present in the atmosphere, this silver mirror can tarnish over time, reducing light output. The source of the sulfur may not be readily obvious. For example, automobile or industrial exhaust can be a source, or even nearby volcanic vents. Likewise, halogens, such as chlorine halogens, can react with silicone lenses, causing them to degrade over time. Fine particulates, such as carbon soot, can deposit onto lenses, reducing the light output. Labs in locations with polluted air should consider filtering systems that clean outside air before it is introduced into the laboratory. In addition, maintenance test chambers should be constructed from components that do not outgas. This includes the MCPCBs that DUTs are mounted on. For example, some soldering fluxes, if left in place, can migrate into the DUT and react with it. Silicone-gap pad material can also release oils that can migrate within the chamber or onto the package.

- **Prevent DUT changes during transport and storage.** DUTs are routinely shuttled from test chambers to centralized optical systems to make required periodic measurements. If the DUTs are left in an uncontrolled environment, they can undergo subtle changes that can slightly alter measurements. For example, DUTs removed from low-humidity high-temperature chambers can absorb moisture during transport and storage. DUTs should be transported in containers than minimize air exchange. If DUTs have to be stored for extended periods (e.g., over a weekend) they should be stored in humidity-controlled enclosures. Samples should also be protected from electrostatic discharge during storage and transport.

- **Correct for reflectivity changes of the sample PCBs.** Elevated temperatures and irradiance, over time, will alter PCBs. Generally, lighter-colored solder masks will darken, and darker-colored solder masks will lighten. These changes will alter the reflectivity of the sample when it is measured. To eliminate this error, a self-absorption correction is applied per **Section 6.1** for each measurement. However, the darkening of thermal adhesive used to attach thermocouples can cause significant near-field absorption issues that may not be fully compensated for by a self-absorption correction.

- **Control emitted radiation in the test chamber.** LEDs and lasers convert a substantial portion of input power to optical radiation. While most devices emit in a Lambertian pattern, device location and light reflections can still result in wide variations in localized irradiance. In the extreme, "hot spots" can develop that elevate some DUT temperatures above the nominal test temperature. **Section 4.4** states that thermal sensors should be shielded. Some thermal adhesives that are typically used for attaching thermocouples will also darken when exposed to excessive radiation. This darkening, over time, can alter thermocouple readings. In addition, test chambers, especially chambers for high-power COB devices, should have a means for absorbing the optical radiation before it is reflected around the chamber.

- **Characterize air temperatures during chamber qualification and/or during pre-test setup.** Test chambers may employ forced convection to control air temperature, or they may maintain air temperature passively (via still air). With both methods, temperature gradients and/or variations can develop that may put DUTs at risk of being stressed at temperatures below the LM-80 limit, or at unnecessarily high temperatures. To prevent this, air temperatures should be characterized using multi-point measurements and/or simulations. This characterization is normally done when the chamber is first qualified, and anytime the thermal load or power density changes significantly. During the characterization, the spatial and temporal variations of air temperature are assessed to ensure that all DUT air temperatures will be above the specified T_A limit and below the desired upper temperature limit.

Annex D – Additional Reading

This Annex is not part of ANSI/IES LM-80-21, Approved Method: Measuring Maintenance of Light Output Characteristics of Solid-State Light Sources; *it is included for informative purposes only.*

Illuminating Engineering Society. ANSI/IES LM-79-19, Approved Method: Electrical and Photometric Measurements of LED Light Sources. New York: IES; 2019.

International Commission on Illumination (CIE). CIE 127:2007, Measurement of LEDs. Vienna: CIE; 2007.

International Commission on Illumination (CIE). CIE 225:2017, Optical Measurement of High-Power LEDs. Vienna: CIE; 2017.

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International Commission on Illumination (CIE). CIE S 025/E:2015, Test Method for LED Lamps, LED Luminaires and LED Modules. Vienna: CIE; 2015.

Joint Electron Device Engineering Council (JEDEC) Solid State Technology Association. JESD 51-14, Transient Dual Interface Test Method for the Measurement of the Thermal Resistance Junction to Case of Semiconductor Devices with Heat Flow through a Single Path. Arlington, VA: JEDEC; 2010.

Joint Electron Device Engineering Council (JEDEC) Solid State Technology Association. JESD51-51, Implementation of the Electrical Test Method for the Measurement of Real Thermal Resistance and Impedance of Light-Emitting Diodes with Exposed Cooling Surface. Arlington, VA: JEDEC; 2012 Apr.

Rencz M, Poppe A, Kollár E, Ress S, Székely V. Increasing the accuracy of structure function based thermal material parameter measurements. IEEE Trans Components Packaging Techn. 2005;28(1).

Székely V, Van Bien T. Fine structure of heat flow path in semiconductor devices: A measurement and identification method. Solid-State El. 1988;31(9):1363-68.

INFORMATIVE REFERENCES

1 Illuminating Engineering Society. ANSI/IES LM-85-20, Approved Method: Optical and Electrical Measurements of LED Packages and Arrays. New York: IES; 2020.

2 Illuminating Engineering Society. ANSI/IES TM-21-21, Approved Method: Projecting Long-Term Lumen, Photon, and Radiant Flux Maintenance of LED Light Sources. New York: IES; 2021.

3 Illuminating Engineering Society. ANSI/IES TM-28-20, Projecting Long-Term Luminous Flux Maintenance of LED Lamps and Luminaires. New York: IES; 2020.

4 Illuminating Engineering Society. ANSI/IES TM-35-19, Technical Memorandum: Projecting Long-Term Chromaticity Coordinate Shift of LED Packages, Arrays, and Modules. New York: IES; 2019.

Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

a) Proposed change accepted for public review without modification
b) Proposed change accepted for public review with modification
c) Proposed change accepted for further study
d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: ______________________________

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Submitter's signature: ______________________________ Date: ______________

2. Title of publications and year published ______________________________
3. Clause (section), sub-clause or paragraph number; and page number: ______________________________
4. My proposal (check one):

   [ ] Change to read as follows

   [ ] Delete and substitute as follows

   [ ] Add new text as follows

   [ ] Delete without substitution

Use underscore to show material to be added (<u>added</u>) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

____

____

____

6. Reason and substantiation:

____

____

____

Select as applicable:

[ ] Additional pages are attached. Number of additional pages:

[ ] Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:

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